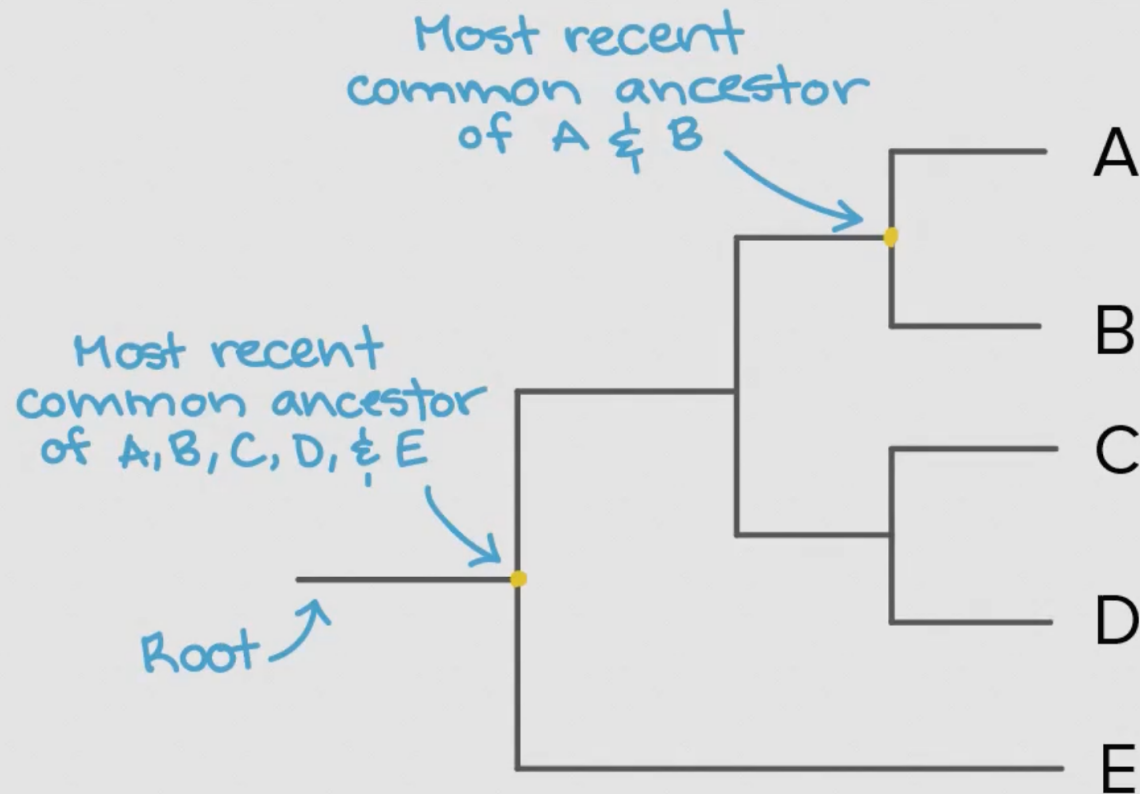


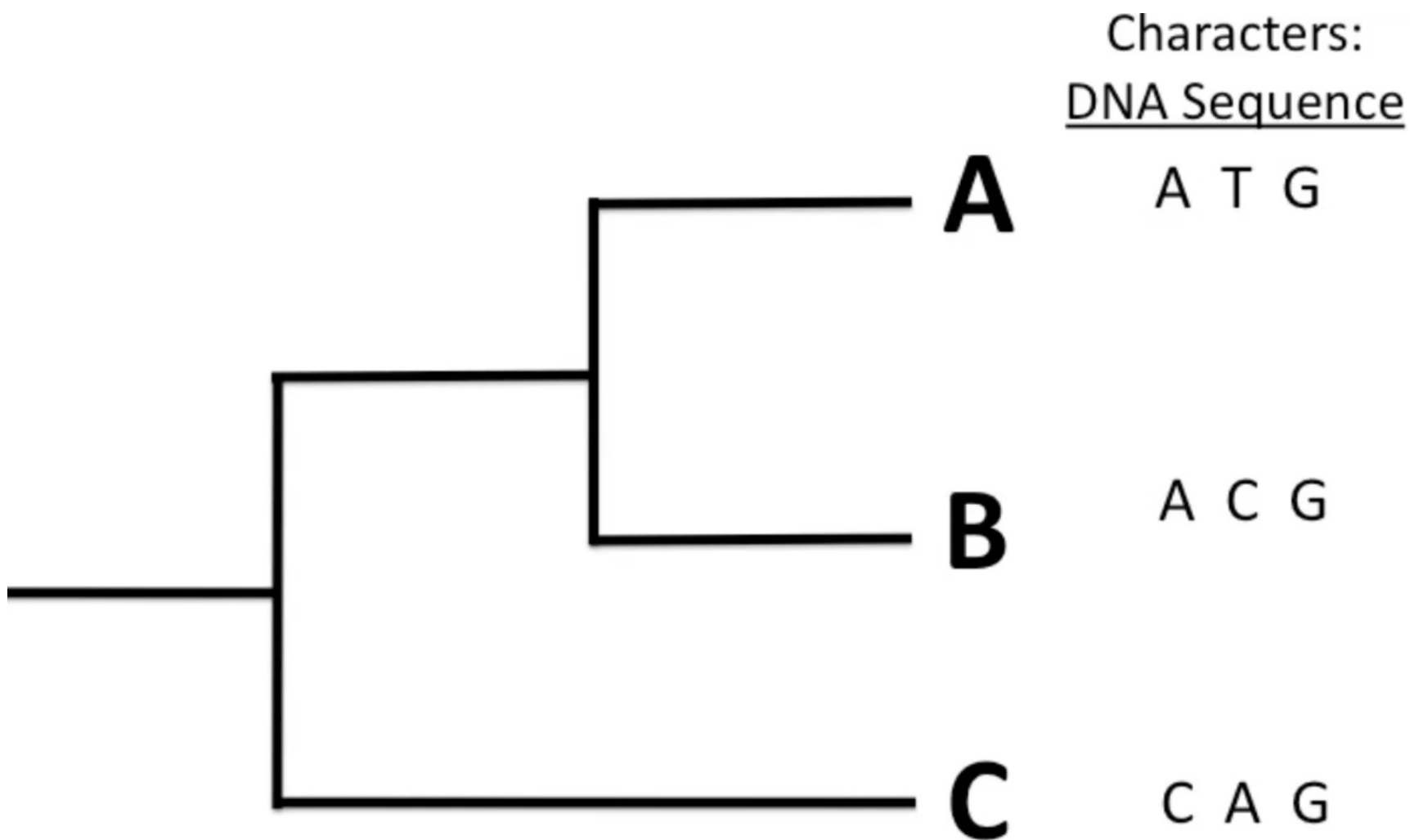
Infectious Disease 'Omics



Introduction to phylodynamics and
populations



ANCESTORS —————> PRESENT-DAY SPECIES



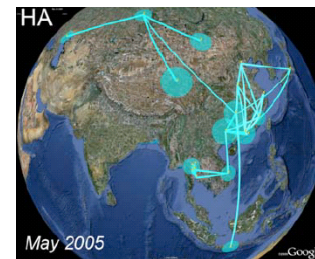
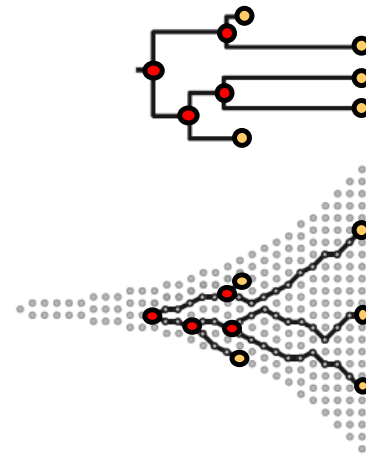
Lecture Overview

1) Introduction

2) Molecular Dating

3) Population Dynamics

4) Phylogeography



Introduction

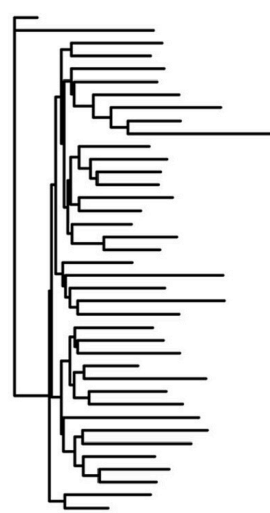
Unifying epidemiology and evolution



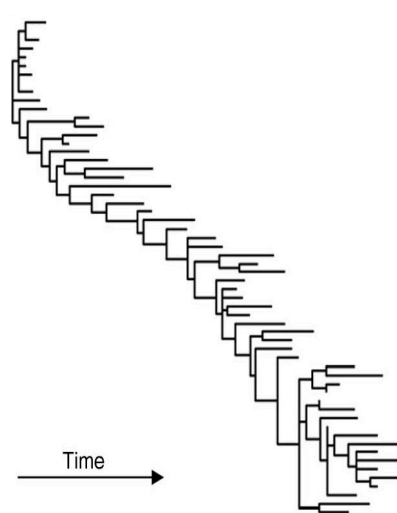
Measles virus
population phylogeny



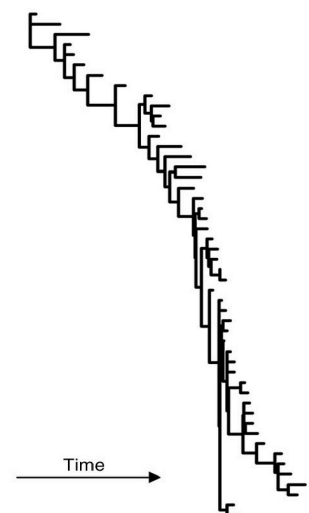
Dengue virus
population phylogeny



HIV
population phylogeny



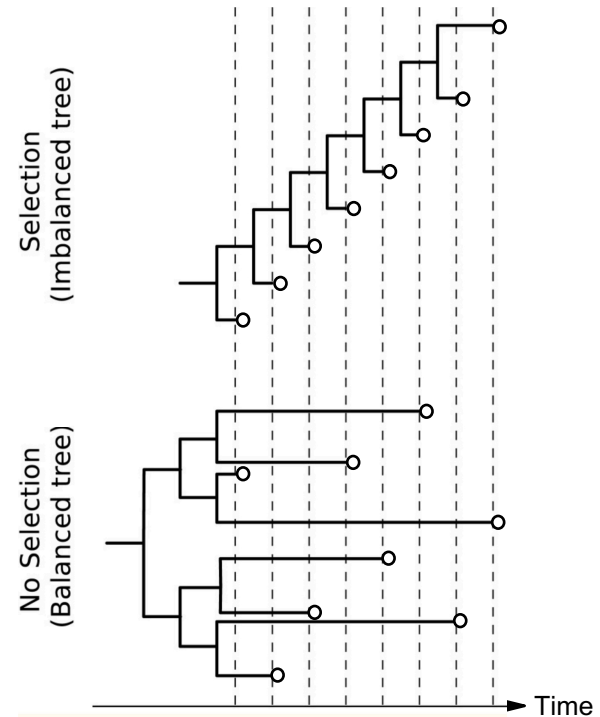
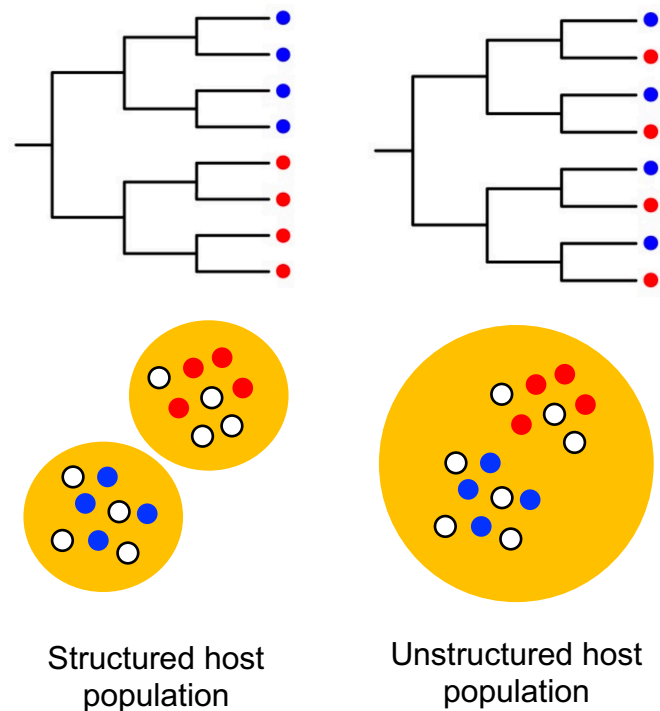
HIV
within host phylogeny



Influenza A virus
population phylogeny

Unifying epidemiology and evolution

- **Phylodynamics:** the study of how epidemiological, immunological and evolutionary processes shape viral phylogenies

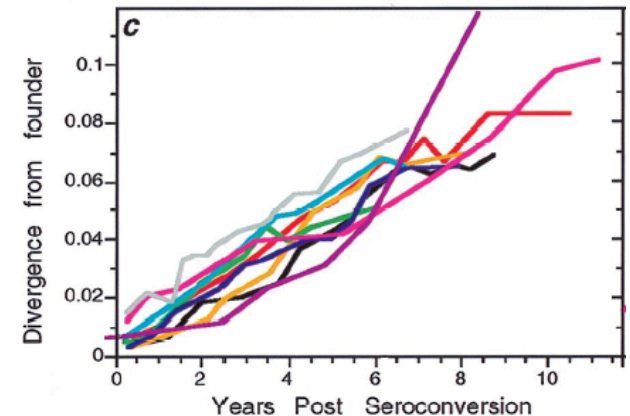


Molecular Dating

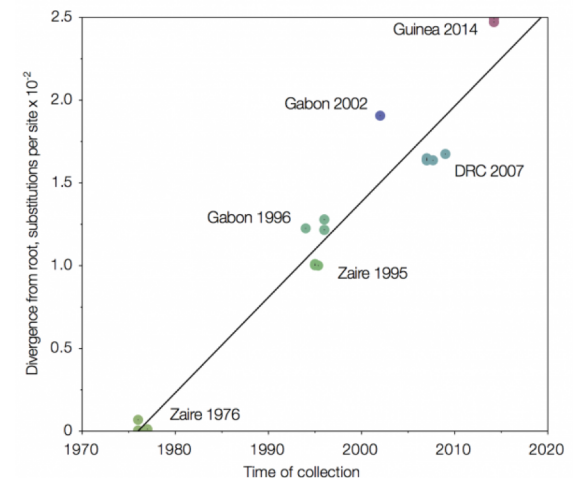
The Molecular Clock Hypothesis

- For a given organism, gene or strain, mutations are fixed at a (nearly) **constant pace**
- Molecular Clock models define a relationship between genetic distance and time:
 - **Strict clock:** all lineages evolve at the **same rate**
 - **Relaxed clock:** Allows evolutionary rates to **change** over time and along lineages

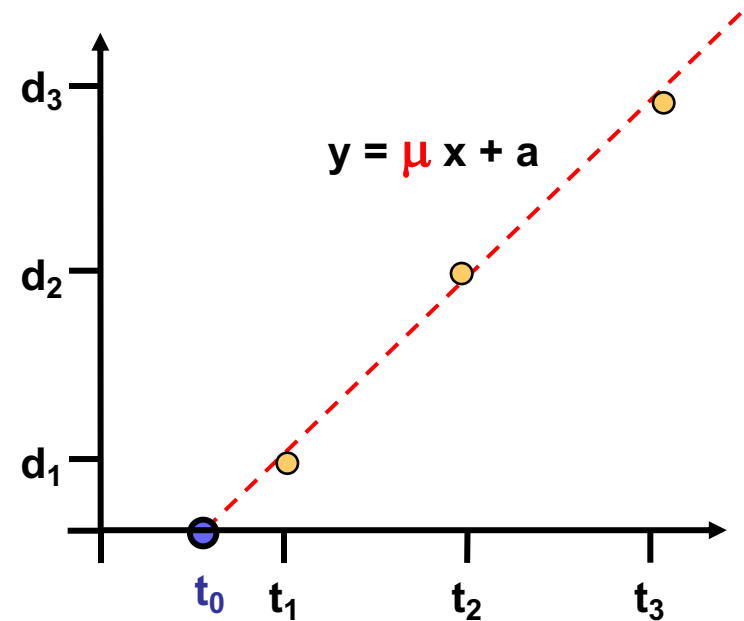
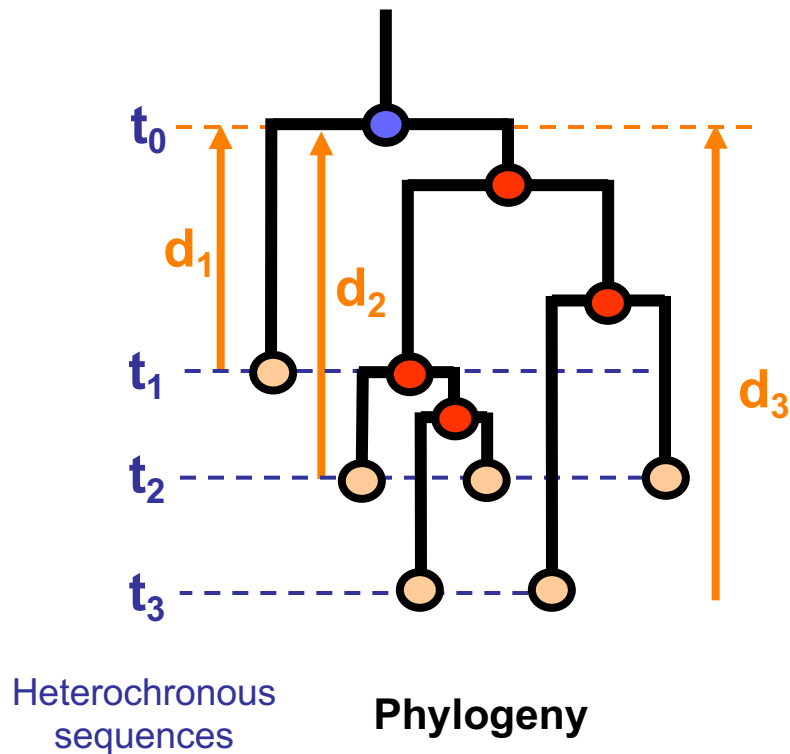
HIV intra-host evolution



Ebolavirus between host evolution



Estimating Substitution Rates



Root-to-Tip linear regression

d_i : genetic distance between sequence and root

t_i : Sampling time of sequence i

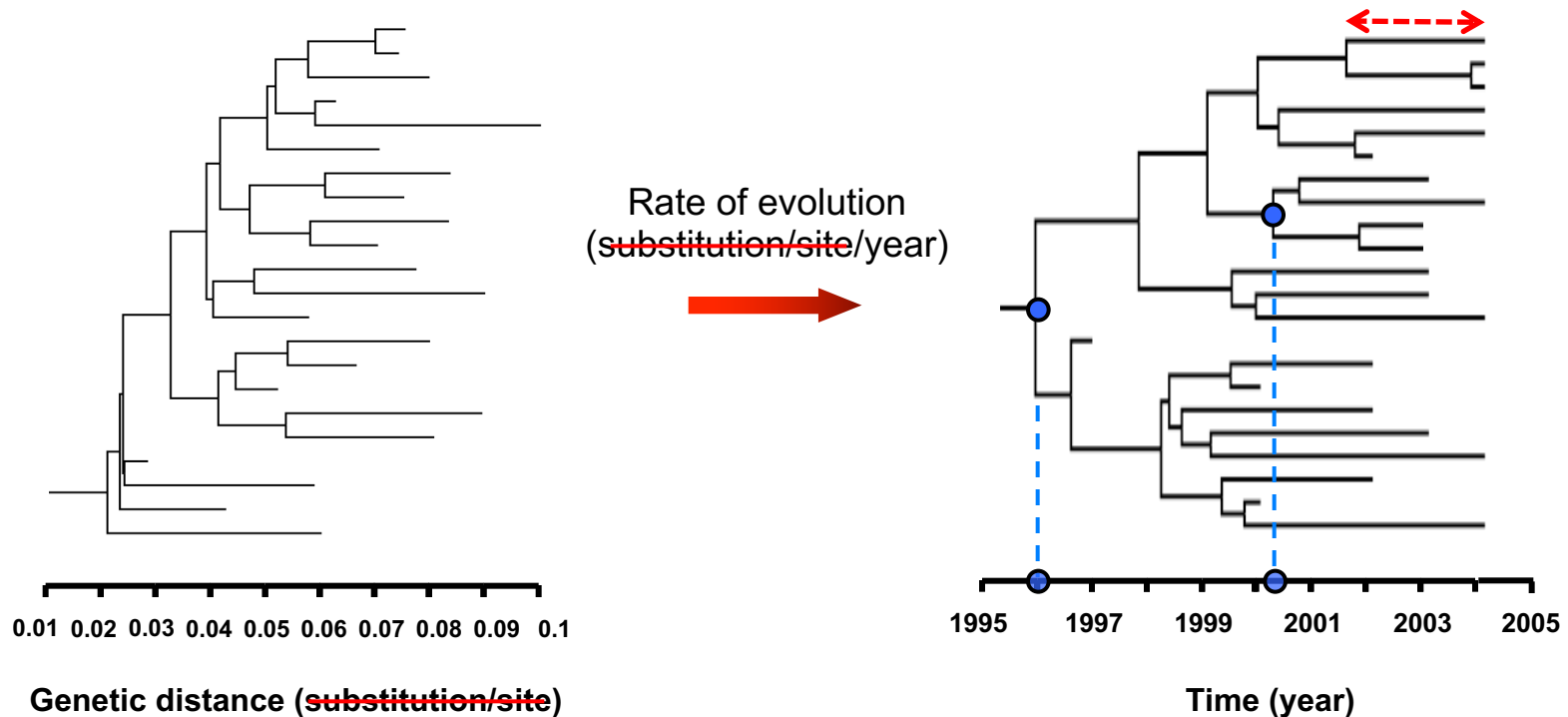
Evolution rate:

$$\mu = \Delta d / \Delta t$$

Time of origin:

$$t_0$$

From Genetic Diversity to Time



- Date the **origin** of an epidemic (root) or outbreak (internal node)
- Estimate transmission **intervals** (internal branch lengths)

Statistical Inference of Evolution Rates

- Root-to-Tip linear regressions rely on two **unrealistic** assumptions:
 - **Constant** evolution rate over time
 - **Independence** of evolutionary histories (violated in internal branches)

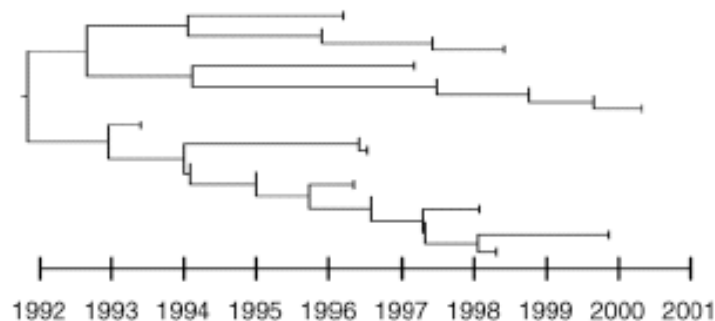
⇒ Incorporation of phylogenetic structure & **statistical hypothesis testing**:

- 1) Each tip of the tree has a known time
- 2) Internal nodes are given arbitrary starting times consistent with their order in the tree
- 3) Vector of internal nodes times ($t_0, t_1, \dots t_n$) and **substitution rate** μ are optimized until the values providing the **best fit** are found

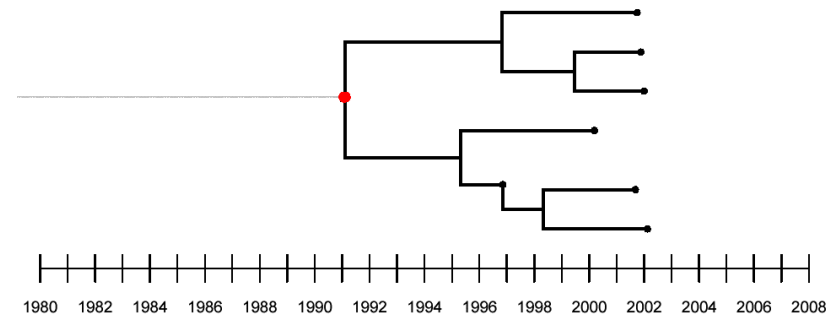
⇒ **Maximum likelihood** or **Bayesian** statistical framework

HIV Sexual Transmission in the UK

Sex between men (Subtype B)



Heterosexual contact (Non-B Subtypes)



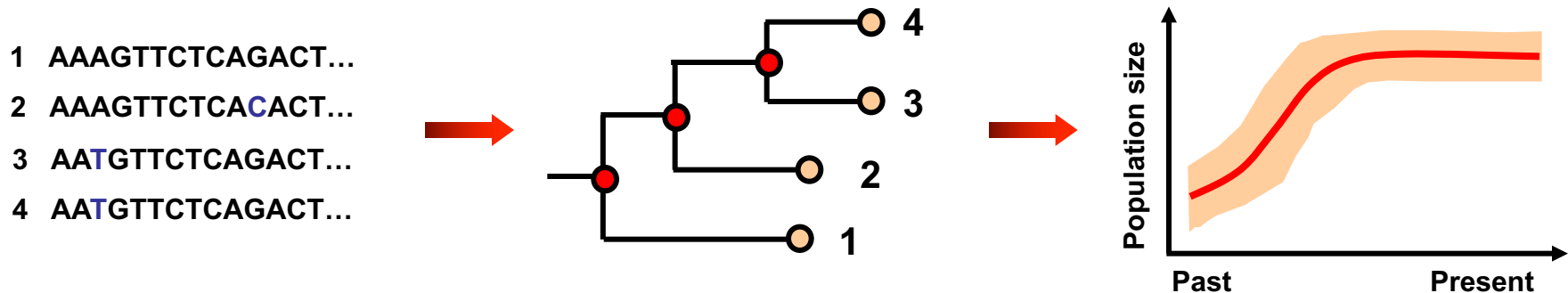
Subtype	n Patients	Frequency of linkage (> 10 individuals)	Median transmission interval*	Frequency of transmission within 6 months post-infection
B	2 126	25%	14 months	25%
Non-B	11 071	5%	27 months	2%

* Median internode interval within a cluster

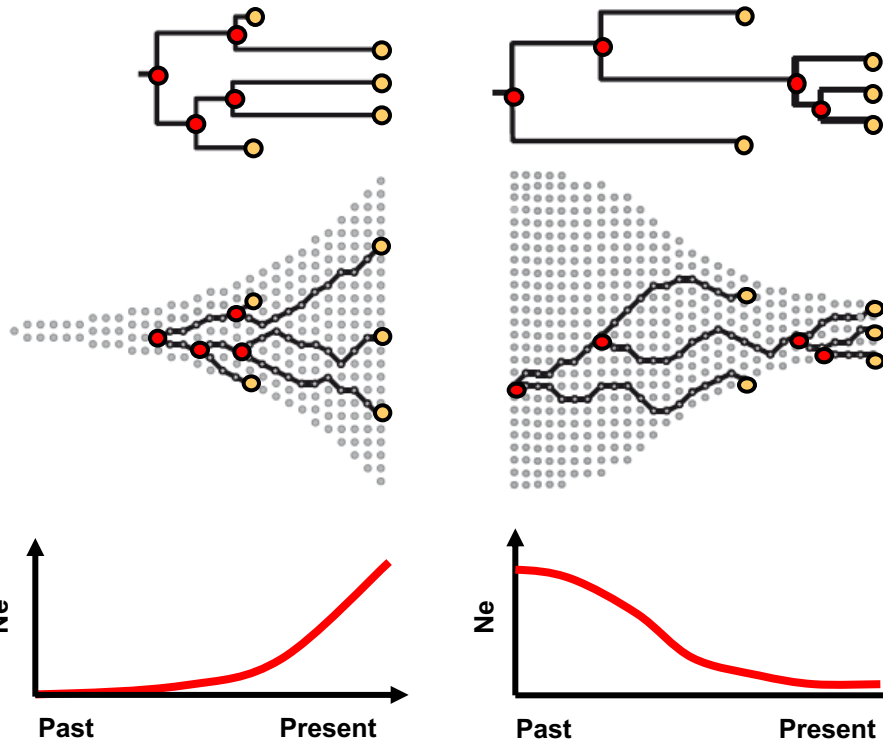
Population Dynamics

Phylogenies & Population Dynamics

- Inference of **demographic characteristics** from phylogenies:
 - **Effective population N_e** (fraction of the total population contributing to the next generation)
 - **Growth rate** of a population
 - **Time of origin** of an epidemic



The Coalescent Theory

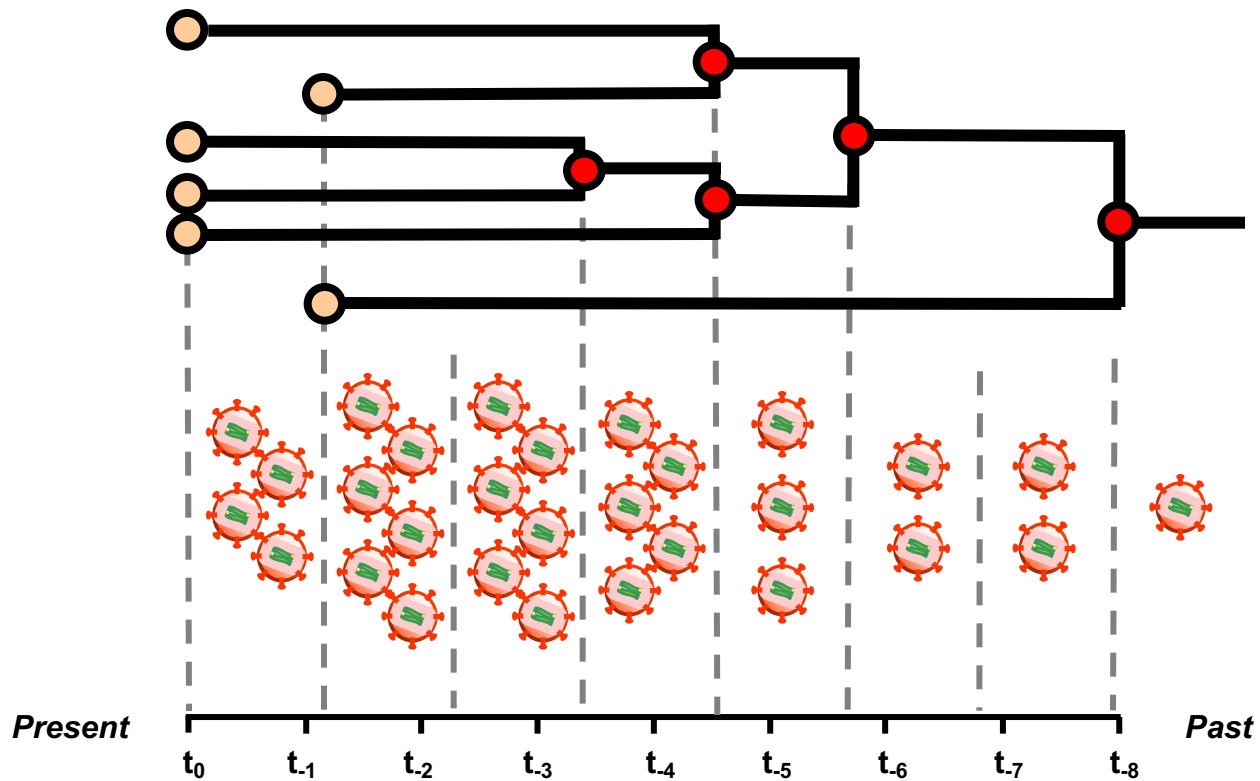


Relationship between genetic diversity and population size:

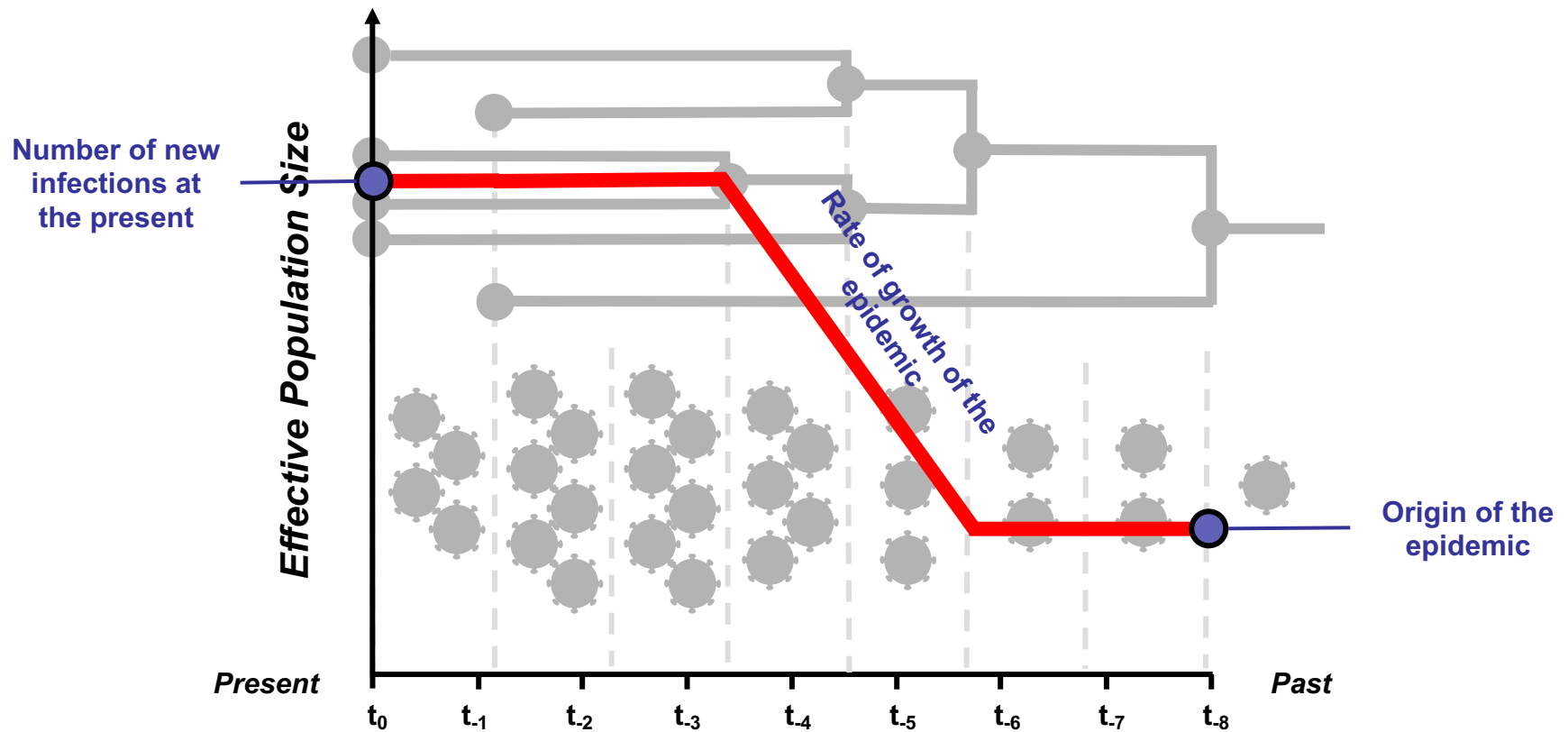
The rate at which lineages 'coalesce' depends on **population size** and **population structure**

Skyline plot: Effective population time over time

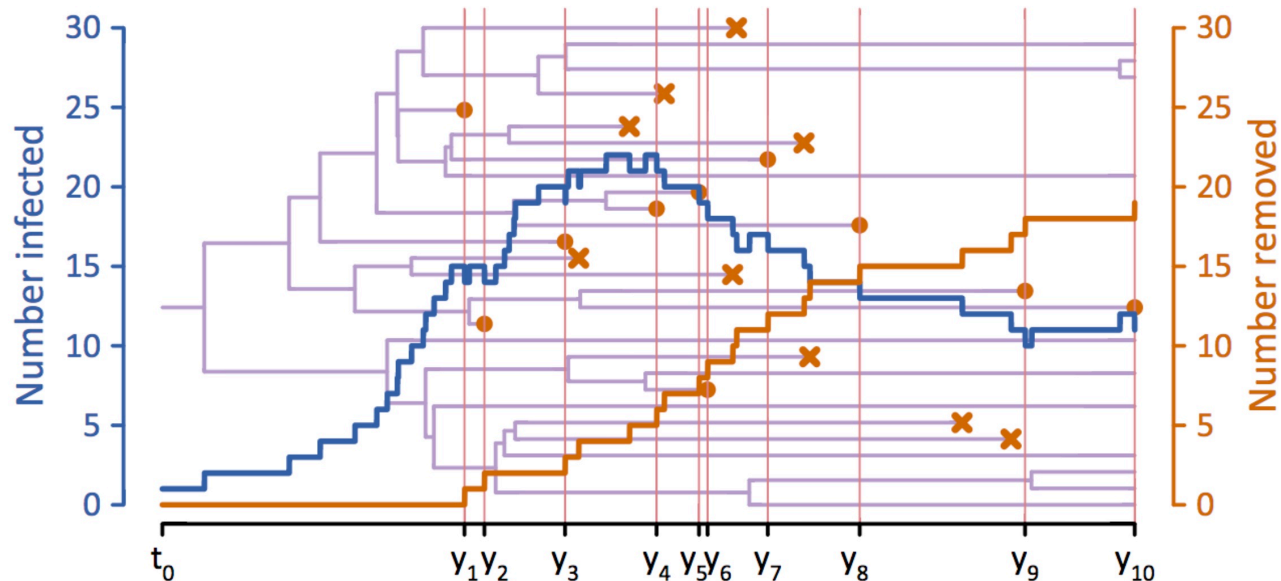
The Coalescent Theory II



The Coalescent Theory II

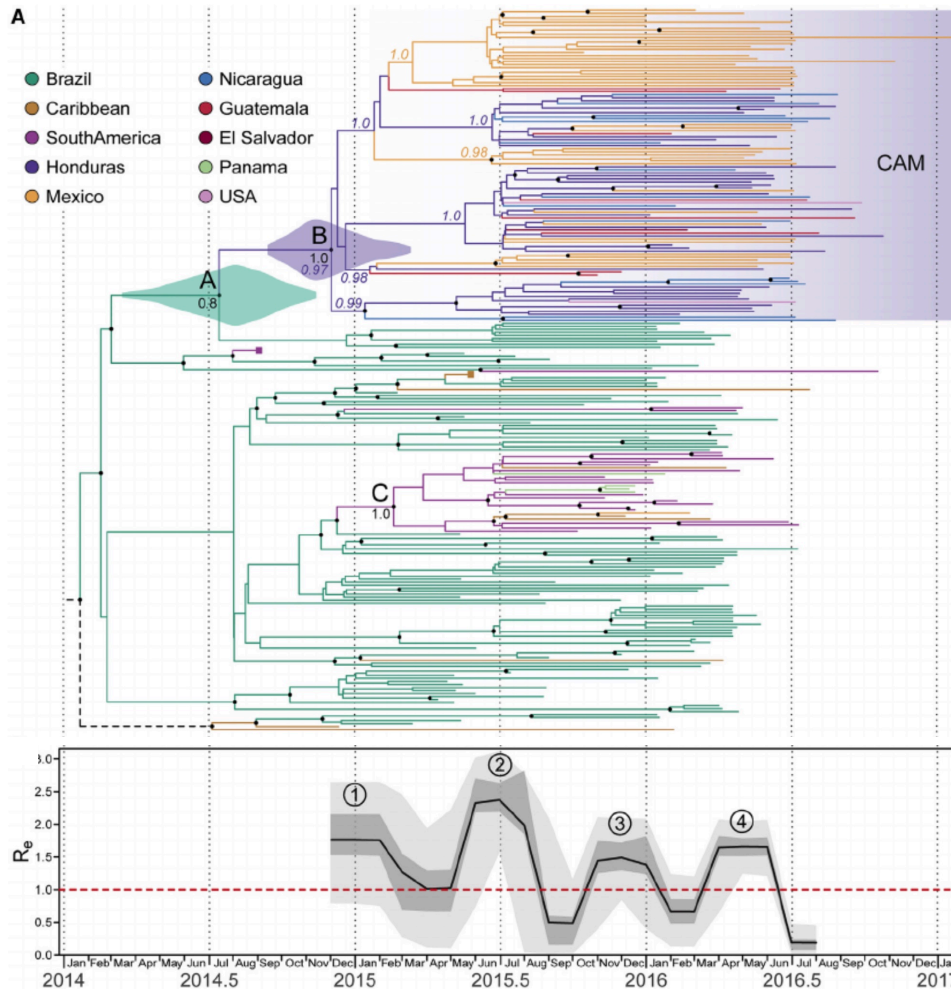


Estimating epidemiological parameters



Number of infectious and removed (i.e., non-infectious) individuals of an epidemic estimated from a viral phylogeny

Estimating Zika Re in Central America & Mexico



Zika **effective reproductive number** (R_e) through time, estimated using a birth-death skyline approach

- (1) ZIKV spread from Honduras to other CA countries
- (2) Rapid radiation of ZIKV lineages
- (3) Period prior to rapid increase in reported ZIKV cases
- (4) Observed epidemic (April–July 2016)

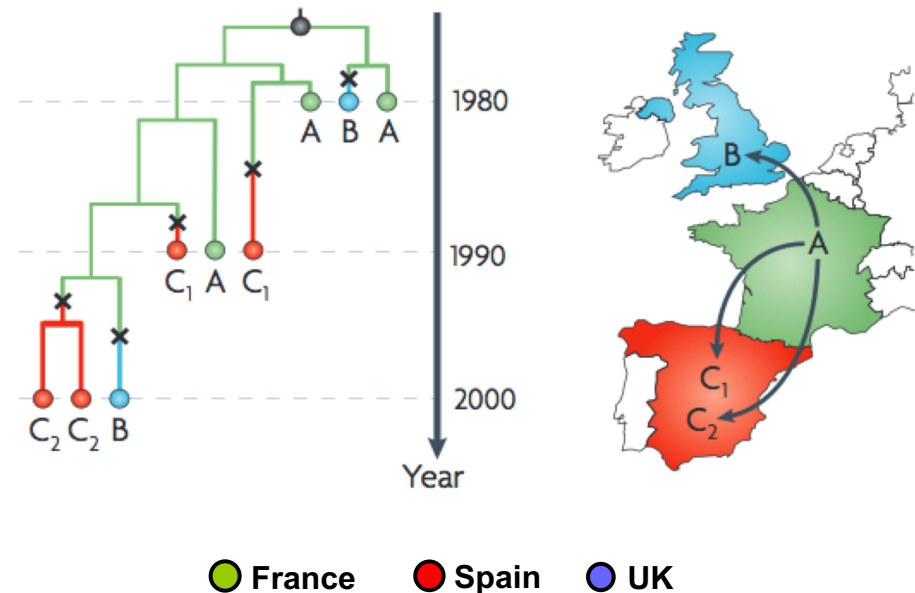
5 – Phylogeography

Reconstructing Geographic Dispersal

- Statistical methods to reconstruct the history of **pathogen spread**:

Geographic location of the pathogen estimated on each branch of a phylogeny

- Parsimony approach:** Ancestral state reconstruction with minimal set of migration events consistent with the phylogeny (Ockham's razor)



Reconstructing Geographic Dispersal

- Parsimony dismisses phylogenetic **uncertainty** and **unequal** state exchange probability
- **Bayesian MCMC** approach:
 - **Ancestral reconstruction** of spatial states along a dated phylogeny:
 - ⇒ Spatial mapping in natural time scales
 - Calculate the **probability** of each ancestral state given the observed data
 - Calibrated under strict / relaxed **molecular clock** + **demographic** models

Constructing Phylogenetic Trees with Bayesian Statistics & MCMC

- **Define Evolutionary Model & Priors**

- Choose substitution and evolutionary rate models.
- Assign prior distributions based on existing knowledge.

1. Initialisation

- **Calculate Likelihood of Observed Genetic Data**

- Determine how well each tree explains the sequence data.

- **Combine Prior and Likelihood to Get Posterior**

- Apply Bayes' theorem to integrate priors and data into posterior probabilities.

- **Sample Trees Using MCMC Algorithms**

- Use Markov Chain Monte Carlo (MCMC) to explore tree-space efficiently.

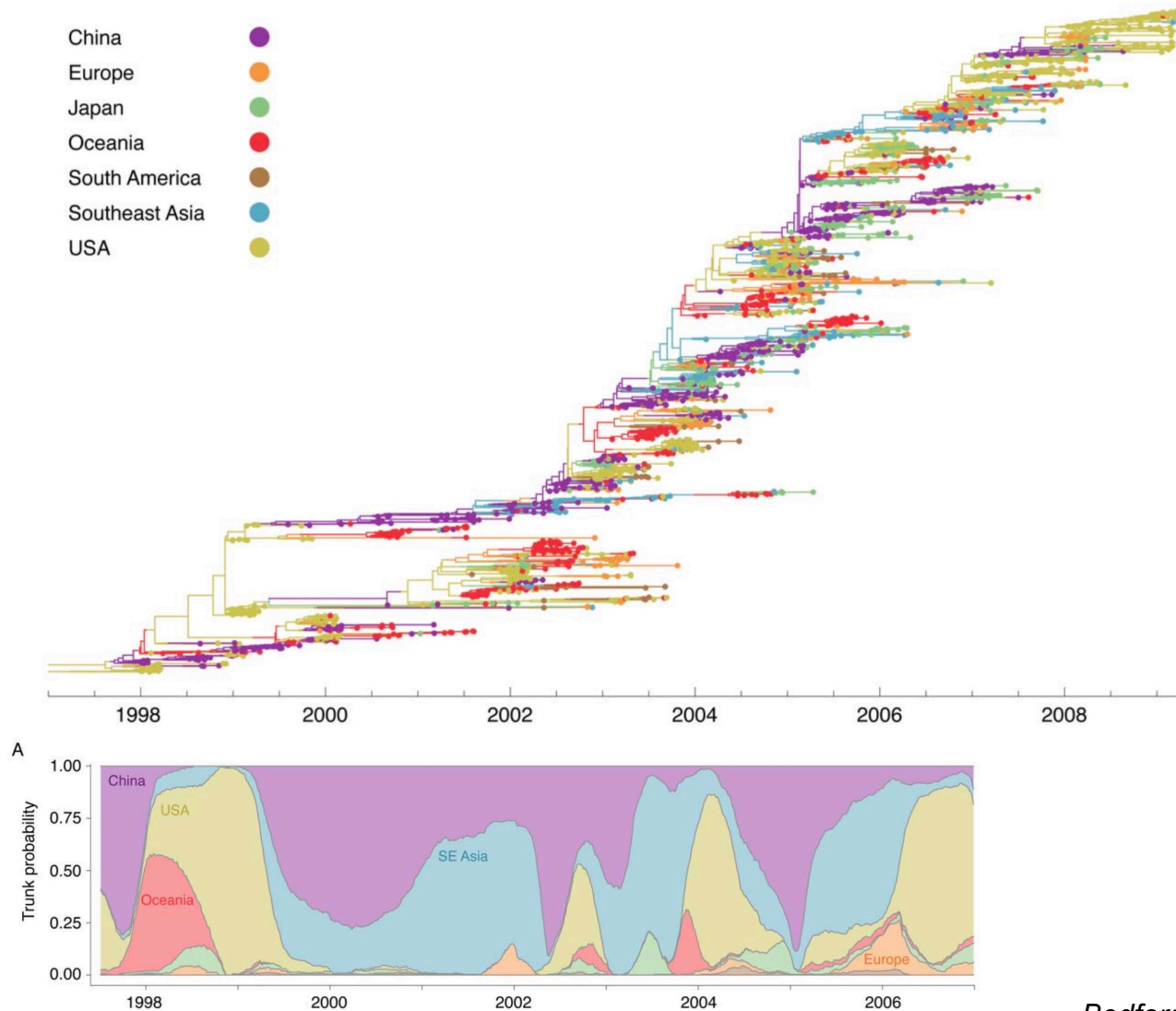
2. Iterative sampling
(Repeated cycles)

- **Summarize Posterior Distribution**

- Identify consensus trees and credible intervals to quantify uncertainty.

3. Convergence

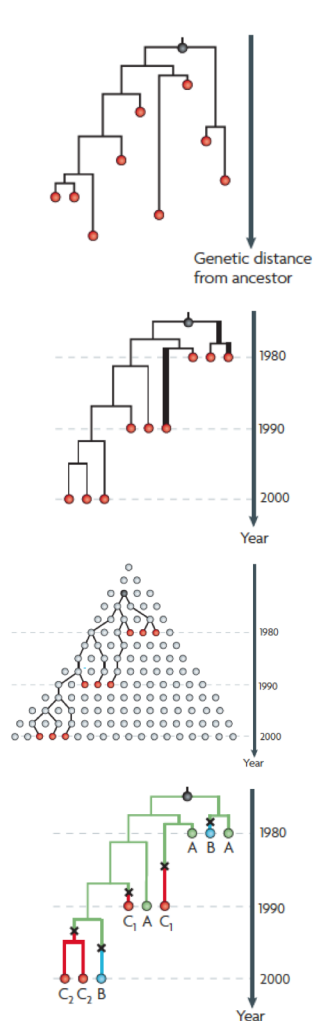
Global migration of Human Influenza A H3N2



Summary

AAGCTTCTCCAGAA
 AAGCATCACAGGATGA
 AAGCTTCAAGGAGGA
 AAGCTACACAAGATGA
 AAGCAACAAGAAAGA

Pathogen Gene Sequences



Evolutionary Tree

Dated Evolutionary Tree

Population Dynamics

Geographic Dispersal

Model of Nucleotide Substitution

Molecular Clock Model

Demographic Model

Translocation Model

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Acknowledgements: Stephane Hue